

Purpose Review Report

Review that README and User Guide are Coherent and Complete

1. Introduction

1.1 Purpose

This document records the formal review of a set of project files.

1.2 Scope

This is the `Purpose` review-set for the `CanvasNet` repository (the repository's single, hierarchy-wide "north star" review per `reviewmark-usage.md`). It proves that the user-facing documentation — `README.md` and the User Guide (`docs/user_guide/introduction.md`) — is internally coherent and complete, and that it accurately and completely describes the library's actual, current public capabilities, including the PDF-rendering system (`DemaConsulting.CanvasNet.Pdf`) and embedded-font subsystem added on the `feature/pdf-document-support` branch.

This is the **second** formal review of this review-set. The first review (fingerprint `b76be037bfealc603ebd0e130abbf5d947f16bb88c7cf30bb27df0f063b6cd2a`, result **Fail**) identified findings F1-F5 below. A subsequent "consolidated documentation-sync pass" (commit `bc86abc`) was made against those findings and others across the repository. This review re-examines the review-set **from scratch**, against the current state of both files, rather than relying on the prior run's cached conclusions.

1.3 Outcomes

Each check must be recorded with one of the following outcomes:

Outcome	Meaning
Pass	The check was performed and the criterion is satisfied
Fail	The check was performed and the criterion is not satisfied
N/A	The check does not apply; justification is required

1.4 Review Details

Field	Value
Project	CanvasNet
Review ID	Purpose
Review Title	Review that README and User Guide are Coherent and Complete
Fingerprint	68d249b4b3058984530ac2db11ecd724861ddf28c95977b847a0d1f74b4cb065
Review Date	2026-10-02
Reviewed Commit	bc86abc1560df73aa141dde47fba8ccbd0f9a2b7
Supersedes	Fingerprint b76be037bfealc603ebd0e130abbf5d947f16bb88c7cf30bb27df0f063b6cd2a (2026-10-02, Fail)

1.5 Reviewers

Name	Role	Organization	Signature	Date
Formal Review Agent (Copilot CLI)	Automated Formal Reviewer	DEMA Consulting	(automated — no signature)	2026-10-02

1.6 Files Under Review

File
README.md
docs/user_guide/introduction.md

2. Review Checklist

2.1 Requirements Checks

Applicable: No

Reason: The review-set contains no requirements files (`README.md` and the user guide are user-facing technical documentation, not requirements artifacts).

2.2 Design Documentation Checks

Applicable: No

Reason: The review-set contains no design documentation files. `docs/design/introduction.md` is explicitly excluded from this review-set's context in `.reviewmark.yaml` (`"!docs/design/introduction.md"`), by design: the `Purpose` review judges the user-facing promise on its own terms, not against the design documentation that is meant to flow *from* it.

2.3 Technical Documentation Checks

Applicable: Yes

#	Check	Outcome	Justification
DOC-01	Documentation is free of technical inaccuracies	Fail	Finding F1 persists unchanged : the User Guide's <code>PdfDocument.Render</code> exceptions list still states that font subtypes <code>/Type0</code> , <code>/Type1</code> , and <code>/Type3</code> are unsupported, contradicting both the current implementation (all three are supported; only <code>/MMType1</code> fails closed) and the User Guide's own preceding prose for the same class. The commit (<code>bc86abc</code>) that fixed Findings F2/F3 did not touch this line.
DOC-02	Documentation is consistent with the current implementation and requirements	Fail	See Findings F1 (persists) and F6 (new): the font-subtype exceptions text (F1) and the complete absence of <code>TilePaint / PathFiller.Fill(..., TilePaint, ...)</code> documentation (F6) leave the documentation inconsistent with, or silent about, implemented, README-advertised public functionality. Findings F2 and F3 from the prior review are now resolved.
DOC-03	All referenced external documents and dependencies are correctly identified	Pass	README's bundled-font license section references <code>src/DemaConsulting.CanvasNet/Fonts/BundledFonts/OFL.txt</code> , <code>.../BundledFonts/README.md</code> , and <code>.../BundledFonts/NotoFonts-OFL.txt</code> ; all three files exist at the stated paths and the attribution (Red Hat Liberation fonts / Noto Project, both SIL OFL 1.1) is correctly stated. Unchanged from the prior review.
DOC-04	Documentation is free of spelling and grammar errors	Pass	<code>npx cspell README.md docs/user_guide/introduction.md</code> reports 0 issues.

2.4 Code Checks

Applicable: No

Reason: The review-set contains no source code files.

2.5 Logic Error Checks

Applicable: No

Reason: The review-set contains no source code files.

2.6 Error Handling & Logging Checks

Applicable: No

Reason: The review-set contains no source code files.

2.7 Usability / Accessibility Checks

Applicable: Yes

Rationale: Both files exist specifically to make the library's public API usable; their completeness and clarity for that purpose is directly in scope.

#	Check	Outcome	Justification
USE-01	The feature or API is easy to use correctly	Pass	Both files provide clear, minimal, compilable-looking code samples for every major feature, including a dedicated "Handling Unsupported Features" example. Unchanged from the prior review.
USE-02	All public APIs are well documented	Fail	Findings F2 and F3 from the prior review are resolved : <code>SystemFontCatalog</code> now has a full API Reference subsection, and <code>TrueTypeFont</code> 's <code>IsBold</code> / <code>IsItalic</code> / <code>IsFixedPitch</code> / <code>GetNameInfo()</code> / <code>FontNameInfo</code> are now documented. However, this check still Fails on new Finding F6: the public <code>TilePaint</code> class and the public <code>PathFiller.Fill(Surface, Path, TilePaint, FillRule, float)</code> overload have no User Guide documentation at all, despite README now advertising "Tile Paint" as a headline feature (added in the very same sync commit that otherwise fixed F2/F3).

2.8 Test Checks

Applicable: No

Reason: The review-set contains no test code files.

2.9 Security Checks

Applicable: No

Reason: The review-set contains no source code files, and neither file makes security claims requiring independent verification beyond the documentation-accuracy checks already covered under section 2.3.

2.10 Code Readability Checks

Applicable: No

Reason: The review-set contains no source code files.

2.11 Requirements vs Documentation Checks

Applicable: No

Reason: The review-set contains no requirements files.

2.12 Requirements vs Implementation Checks

Applicable: No

Reason: The review-set contains no requirements files or source code files.

2.13 Requirements vs Testing Checks

Applicable: No

Reason: The review-set contains no requirements files or test code files.

2.14 Code vs Design Documentation Checks

Applicable: No

Reason: The review-set contains no source code files or design documentation files.

3. Conclusion

3.1 Summary of Findings

#	Status	Check	Finding
F1	Open (unchanged)	DOC-01	High confidence. docs/user_guide/introduction.md , in the PdfDocument.Render(int, int, int) Exceptions list (around line 1242), still states: " <i>UnsupportedImageFeatureException : Thrown for a well-formed but unsupported color space, stream filter, Form XObject, font subtype (/Type0 / /Type1 / /MMType1 / /Type3), ...</i>" — implying /Type0 , /Type1 , and /Type3 fonts are all unsupported. This contradicts: (a) the implementation (src/DemaConsulting.CanvasNet.Pdf/PdfDocument.Fonts.cs 's ResolveFont /dispatch, which resolves TrueType / Type1 → BuildResolvedSimpleFont , Type0 → BuildResolvedCompositeFont , Type3 → BuildResolvedType3Font , and throws only for any <i>other</i> subtype, e.g. MMType1); and (b) the same document's own preceding "Documented scope boundaries" sentence (just above, around line 1125), which correctly lists only /MMType1 fonts (among font-related items) as out of scope. Confirmed via git show bc86abc -- docs/user_guide/introduction.md that this specific line was not touched by the sync pass, even though the pass's own commit message claims font-subtype "unsupported" claims were corrected. Recommendation: Change the Exceptions bullet to read "font subtype (/MMType1)" (or otherwise state that only /MMType1 remains unsupported among simple/composite/procedural font subtypes), matching the surrounding prose.
F2	Resolved	USE-02	Fonts.SystemFontCatalog now has a complete ### SystemFontCatalog API Reference subsection in the User Guide (Fonts , FindBestMatch , LoadBundledFallback , SystemFontInfo all documented), added in commit bc86abc . No further action needed.
F3	Resolved	USE-02	TrueTypeFont 's IsBold / IsItalic / IsFixedPitch properties and GetNameInfo() / FontNameInfo record are now listed in the User Guide's TrueTypeFont code block and prose, added in commit bc86abc . No further action needed.
F4	Open (unchanged) — Observation	—	Medium confidence. README.md 's " PDF Document" feature bullet still does not mention password-protected/encrypted-PDF support, even though PdfDocument.Open(..., string? password = null) is public API and the User Guide documents RC4/AES-128/AES-256 decryption in detail. Not a contradiction, but a completeness gap for a substantial, headline-worthy capability. Recommendation: Add a short clause to the README's PDF feature bullet, e.g. "... and transparently opens common password-protected documents."

#	Status	Check	Finding
F5	Open (unchanged) — Observation	—	Low confidence. PDF rendering still has no dedicated numbered "Example" in the User Guide's Examples section (Examples 1-13 cover every other major feature), only the brief inline snippet inside the PdfDocument API Reference subsection. Recommendation: Consider adding "Example 14: Rendering a PDF Page" for parity.
F6	New	DOC-02 / USE-02	High confidence. TilePaint (src/DemaConsulting.CanvasNet/Drawing/TilePaint.cs) is a public sealed class (constructor, Surface / Transform / XStep / YStep properties, WithTransform), and PathFiller (src/DemaConsulting.CanvasNet/Drawing/PathFiller.cs) has a public Fill(Surface, Path, TilePaint, FillRule, float) overload — the tiling-pattern analogue of the already-documented Gradient overload. Commit bc86abc added a brand-new " Tile Paint " bullet to README.md 's Features list advertising this capability, and its own commit message claims to "document the Fill(Surface, Path, TilePaint, FillRule, float) overload" — but that documentation was added only to the design docs (docs/design/...), not to the User Guide. A full-text search of docs/user_guide/introduction.md for "Tile"/"TilePaint" finds zero matches outside of unrelated PDF tiling-pattern prose; there is no ### TilePaint API Reference subsection, and the #### PathFiller Methods subsection lists only the solid-color and Gradient overloads, omitting the TilePaint overload entirely. A reader following the User Guide alone has no way to discover or use this now README-advertised feature. Recommendation: Add a ### TilePaint API Reference subsection (constructor, properties, WithTransform) and document the third PathFiller.Fill(Surface, Path, TilePaint, FillRule, float) overload in #### PathFiller Methods , mirroring the existing Gradient treatment.
F7	New — Observation	—	Low confidence. README.md 's " PDF Document" feature bullet says rendering covers "... TrueType text ..." only, which undersells the capability: the User Guide (and implementation) show PdfDocument.Render actually supports TrueType , Type1 , Type0 (composite/CID), and Type3 (procedure-painted) font subtypes, plus Symbol/ZapfDingbats fallback. This is a wording simplification rather than a contradiction, but it is incomplete relative to what the User Guide itself documents in depth. Recommendation: Consider broadening the wording to "... and text using any of the PDF simple/composite/procedural font subtypes it supports (substituting a matching system or bundled fallback font when none is embedded)," or similar.

3.2 Overall Outcome

Overall Outcome: Fail

Two of the three prior Fail-driving findings (F2, F3) are confirmed resolved by commit `bc86abc`. However, the review-set still Fails:

- **F1 persists unchanged** — the exact same self-contradictory, verifiably-wrong `PdfDocument.Render` font-subtype exceptions claim identified in the first review is still present, untouched by the sync pass.
- **F6 is a new instance of the same defect pattern as F2/F3** — a public, README-advertised headline feature (`TilePaint`, introduced in this very sync pass) has zero User Guide API documentation, exactly the coherence gap this review-set exists to catch. Notably, the sync commit's own message claims to have documented the `TilePaint` `Fill` overload, but did so only in the design docs — not in the (in-scope) User Guide.

Findings F4, F5, and F7 remain lower-severity observations that do not independently drive the Fail outcome but are recorded for completeness.

Required follow-up before this review-set can be considered Pass/approved:

1. Correct the `PdfDocument.Render` Exceptions entry in the User Guide so only `/MMType1` is listed as an unsupported font subtype, matching the surrounding prose and the implementation (F1).
2. Add a `TilePaint` API Reference subsection and document `PathFiller`'s `TilePaint` `Fill` overload in the User Guide (F6).
3. Optionally address F4, F5, and F7 for completeness/parity.

No code or documentation was modified by this review; all findings are reported here for the orchestrator/maintainers to action.